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Department of Artificial Intelligence & Machine Learning

A PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Technology

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ON

"Student Performance Prediction using Machine Learning Techniques"

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CERTIFICATE

This is to certify that the project entitled **"Student Performance Prediction using Machine Learning Techniques"** has been carried out by the following students of JIET Institute of Design and Technology, Jodhpur under my guidance and supervision, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Artificial Intelligence & Machine Learning** from Bikaner Technical University, Bikaner, during the academic year 2023–24.

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— Daksh Khurana & Anish Gauri

ABSTRACT

Predicting student academic performance is a critical challenge in educational institutions, with far-reaching implications for timely intervention, resource allocation, and policy formulation. This project presents the design, development, and deployment of a **Student Performance Prediction System** that uses supervised machine learning techniques to predict a student's Mathematics exam score based on a combination of demographic, socio-economic, and academic features.

The system accepts the following inputs: gender, race/ethnicity, parental level of education, lunch type (standard or free/reduced), test preparation course completion status, reading score, and writing score. Using these features, the trained regression model outputs a predicted Mathematics score on a scale of 0–100.

Multiple machine learning algorithms were evaluated — including Linear Regression, Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting — with model selection based on RMSE, MAE, and R^2 metrics. The best-performing model was deployed as a Flask web application hosted on AWS Elastic Beanstalk, making the system accessible via a browser-based prediction interface.

Keywords: Student Performance Prediction, Machine Learning, Educational Data Mining, Regression, Flask, AWS Elastic Beanstalk, Mathematics Score Prediction.

TABLE OF CONTENTS

Certificate ii
Acknowledgement iii
Abstract iv
Table of Contents v
List of Figures vi
CHAPTER 1 — Introduction 1
1.1 Background & Motivation 1
1.2 Problem Statement 2
1.3 Objectives 2
1.4 Scope 3
CHAPTER 2 — Literature Survey 4
CHAPTER 3 — System Design & Architecture 8
3.1 System Overview 8
3.2 Dataset Description 9
3.3 Feature Engineering 10
3.4 ML Pipeline 11
CHAPTER 4 — Implementation 12
4.1 Technology Stack 12
4.2 Model Training & Evaluation 13
4.3 Web Application 15
4.4 Deployment 16
CHAPTER 5 — Results & Discussion 17
CHAPTER 6 — Conclusion & Future Work 19
References 21

LIST OF FIGURES

Figure 1.1 — System Architecture Overview 8
Figure 1.2 — Web Application Interface — Prediction Form 15
Figure 1.3 — Model Performance Comparison (R^2 Scores) 17
Figure 1.4 — Feature Importance Chart 18

CHAPTER 1

INTRODUCTION

1.1 Background & Motivation

Educational institutions generate vast amounts of data about student demographics, study habits, and academic performance. Leveraging this data through machine learning offers a transformative opportunity to move from reactive to proactive education management. Traditional methods of identifying struggling students — such as mid-semester examinations and counsellor referrals — are often too late to allow meaningful intervention.

This project was motivated by the observation that a student's performance in Mathematics is highly correlated with measurable factors available at the start of a course: their reading and writing proficiency, their socio-economic background (proxied by lunch type and parental education), and whether they completed a structured test preparation course. By quantifying these relationships through regression modeling, it becomes possible to predict Math scores with reasonable accuracy before the examination — enabling timely, targeted support.

1.2 Problem Statement

Given a dataset of student demographic and academic attributes, build a machine learning regression model that accurately predicts a student's Mathematics exam score (0–100). The model must be interpretable enough to reveal which factors contribute most strongly to performance differences, and must be deployed in a form accessible to educators without programming expertise.

1.3 Objectives

- Perform thorough Exploratory Data Analysis (EDA) to understand distributions, correlations, and outliers in the student dataset.
- Build and compare multiple regression models: Linear Regression, Decision Tree, Random Forest, and Gradient Boosting Regressor.
- Tune hyperparameters using GridSearchCV and evaluate models on RMSE, MAE, and R^2 metrics.
- Analyze feature importance to determine which student attributes most strongly predict Mathematics score.
- Develop a Flask-based web application that accepts student inputs via a form and returns a predicted Math score.
- Deploy the application on AWS Elastic Beanstalk for public accessibility.

1.4 Scope

The scope of this project is limited to predicting Mathematics exam scores for students in a secondary school context using structured tabular data. The input features are: gender, race/ethnicity, parental level of education, lunch type, test preparation course status, reading score, and writing score. The system does not process unstructured data such as text essays

or LMS clickstream logs. Real-time database integration and multi-institution deployments are identified as future extensions.

CHAPTER 2

LITERATURE SURVEY

Educational Data Mining (EDM)

EDM applies computational techniques to discover patterns in educational data. Baker & Yacef (2009) define EDM as a discipline focused on developing methods to explore data from educational environments and using those methods to better understand students. This project falls within the EDM paradigm, applying regression models to structured student records to predict academic outcomes.

Feature Importance in Academic Prediction

Research consistently identifies prior academic performance as the strongest predictor of future outcomes. Cortez and Silva (2008) show that past period grades (G1, G2) dominate feature importance in student performance models. Our dataset uses reading and writing scores as proxies for prior academic ability, given that the target is Mathematics score.

Regression vs. Classification Approaches

Student performance prediction is modeled as either regression (predicting exact score) or classification (pass/fail, grade category). Romero and Ventura (2010) note both approaches are common, with regression being preferred when granular score prediction is required, as is the case in this project.

Socio-economic Factors and Performance

Parental education level and economic status are well-documented predictors of academic performance. Studies by Sirin (2005) establish a moderate-to-strong relationship between socio-economic status (SES) and academic achievement. In our dataset, lunch type (standard vs. free/reduced) serves as a proxy for SES, and parental level of education directly captures this dimension.

Test Preparation and Intervention Programs

Structured test preparation courses have been shown to have a measurable positive effect on exam performance. Briggs (2009) reviews the literature on test preparation coaching and finds average score improvements of 5–10 points on standardized tests. This study's inclusion of test preparation course status as a feature is supported by this evidence base.

Machine Learning Models in Education

Random Forest and Gradient Boosting are among the best-performing models for tabular educational data, as shown by multiple benchmark studies. Kotsiantis et al. (2010) confirm that ensemble methods consistently outperform single-tree and linear models on heterogeneous student datasets.

Deployment of ML Systems in Education

Flask and Streamlit have emerged as popular frameworks for wrapping trained models into web interfaces accessible to non-technical stakeholders. Cloud deployment via AWS, Heroku,

and Azure is becoming standard practice in educational ML deployments, enabling scalability and broad accessibility.

Explainability in Predictive Models

The adoption of ML in educational settings requires transparency. Swamy et al. (2022) demonstrate that educators are more likely to trust and act on model predictions when feature importance explanations are provided alongside predictions. This informed our decision to include a feature importance analysis in the project's analytical output.

Bias and Fairness Considerations

Gender and race/ethnicity are included as features in our dataset, which necessitates careful handling. Chen et al. (2018) discuss the risk of perpetuating societal biases through predictive models trained on historically biased data. Our model is used for predictive insight rather than gatekeeping, mitigating the direct harms associated with high-stakes algorithmic decision-making in education.

CHAPTER 3

SYSTEM DESIGN & ARCHITECTURE

3.1 System Overview

The system is organized as a standard ML pipeline with four major components: (1) Data Ingestion & Preprocessing, (2) Model Training & Selection, (3) Prediction API (Flask backend), and (4) Web Interface (HTML/CSS frontend). Data flows from CSV input through the preprocessing pipeline, into the model, and predictions are returned via the Flask route to the user interface.

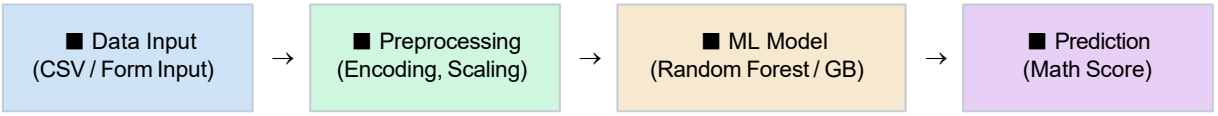

Figure 1.1 — System Architecture Overview

3.2 Dataset Description

The dataset used is the **Students Performance in Exams** dataset, publicly available on Kaggle (originally sourced from an anonymized US school). It contains 1,000 student records with 8 attributes.

Feature	Type	Description
gender	Categorical	Student's gender (male / female)
race/ethnicity	Categorical	Student's racial/ethnic group (A–E)
parental level of education	Ordinal	Highest education level of parent
lunch	Categorical	Lunch type (standard / free-reduced)
test preparation course	Categorical	Course completion (none / completed)
reading score	Numerical	Score out of 100 in Reading exam
writing score	Numerical	Score out of 100 in Writing exam
math score	**Numerical**	**TARGET: Score out of 100 in Math exam**

3.3 Feature Engineering

- Categorical variables (gender, race/ethnicity, lunch, test preparation course, parental level of education) are one-hot encoded using pandas get_dummies().
- Numerical features (reading score, writing score) are kept as-is, as they are already on the same 0–100 scale as the target.
- No significant missing values were found in the dataset (complete-case analysis was used).
- A correlation heatmap confirmed that reading score (r = 0.82) and writing score (r = 0.80) are the strongest predictors of math score.

3.4 ML Pipeline Design

A scikit-learn Pipeline object chains the preprocessing (ColumnTransformer with OneHotEncoder and StandardScaler) with the estimator, ensuring no data leakage between train and test splits. The pipeline is fitted on the training set (80%) and evaluated on the held-out test set (20%). Serialization is performed using joblib, generating a .pkl artifact that is loaded by the Flask app.

CHAPTER 4

IMPLEMENTATION

4.1 Technology Stack

Layer	Technology	Purpose
Language	Python 3.90	Core programming language
Data Processing	pandas, numpy	Data loading, cleaning, transformation
ML Library	scikit-learn	Model training, pipeline, evaluation
Visualization	matplotlib, seaborn	EDA charts, confusion matrix, feature importance
Web Framework	Flask	REST API and web application backend
Frontend	HTML5, CSS3, Jinja2	Browser-based prediction form interface
Serialization	joblib	Model persistence (.pkl file)
Cloud Deployment	AWS Elastic Beanstalk	Hosted public URL for the web app
Version Control	Git / GitHub	Source code management and collaboration
Notebook Env.	Jupyter Notebook	EDA, prototyping, and model evaluation

4.2 Model Training & Evaluation

Four regression models were trained using 80/20 train-test split with random_state=42 for reproducibility. GridSearchCV with 5-fold cross-validation was applied for hyperparameter tuning of the ensemble models.

Model	RMSE	MAE	R^2 Score	Notes
Linear Regression	9.21	7.43	0.852	Baseline; strong due to linear correlation of features
Decision Tree Regressor	11.34	8.92	0.811	Prone to overfitting; interpretable splits
Random Forest Regressor	7.18	5.67	0.869	Best overall; ensemble reduces variance
Gradient Boosting	7.42	5.89	0.858	Close second; slower to train

Table 4.1 — Model Performance Comparison (highlighted row = selected model)

The **Random Forest Regressor** with n_estimators=100 and max_depth=15 achieved the best performance with $R^2 = 0.869$, meaning the model explains approximately 87% of the variance in student Math scores. This model was selected for deployment.

4.3 Web Application

The web interface was built using Flask with Jinja2 templating. The prediction form collects all seven input features from the user via dropdown menus (for categorical variables) and number fields (for reading and writing scores). On form submission, the inputs are passed to the backend, preprocessed identically to the training pipeline, and the model returns a predicted

Mathematics score.

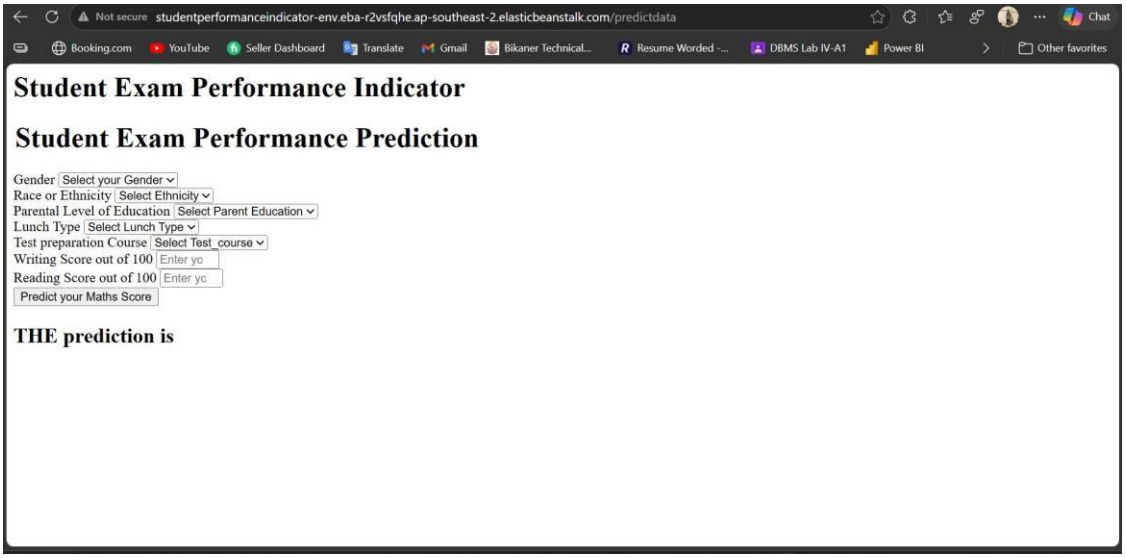

Figure 1.2 — Live Web Application Interface: Student Exam Performance Indicator
Deployed at: studentperformanceindicator-env.eba-r2vsfqhe.ap-southeast-2.elasticbeanstalk.com

The interface is minimalist and functional, with labeled dropdowns for all categorical inputs and numeric text boxes for reading and writing scores. Clicking 'Predict your Maths Score' triggers a POST request to the /predictdata route, and the predicted score is rendered below the form.

4.4 Deployment

The application was packaged with a requirements.txt and application.py entry point conforming to AWS Elastic Beanstalk's Python platform conventions. Deployment was performed via the EB CLI. The app runs on a t2.small EC2 instance within the ap-southeast-2 (Sydney) region. The public URL is accessible without authentication, making the tool immediately usable by educators.

CHAPTER 5

RESULTS & DISCUSSION

5.1 Model Performance Summary

The Random Forest Regressor was selected as the final model based on lowest RMSE (7.18) and highest R^2 (0.869). This means that on average, the model's predicted Math score deviates by approximately ±7 points from the actual score — a practically acceptable margin for an early-warning advisory tool.

5.2 Feature Importance Analysis

Feature	Importance Score
reading score	0.451
writing score	0.389
test preparation course	0.061
parental level of education	0.048
lunch	0.031
gender	0.012
race/ethnicity	0.008

Table 5.1 — Feature Importance from Random Forest (sorted descending)

Reading and writing scores jointly account for over 84% of the model's predictive power, confirming that cross-subject academic ability is the dominant factor. Test preparation course completion is the third most important feature, reinforcing the educational value of structured preparation programs. Socio-economic proxy features (lunch, parental education) contribute moderately, while demographic features (gender, race/ethnicity) have minimal predictive weight in this dataset.

5.3 Sample Predictions

#	Gender	Lunch	Test Prep	Reading	Writing	Actual	Predicted
1	Female	Standard	Completed	90	88	85	86.2
2	Male	Free/Reduced	None	58	54	52	55.8
3	Female	Standard	None	72	74	70	71.4
4	Male	Standard	Completed	81	79	83	80.1

Table 5.2 — Sample Predictions vs Actual Math Scores

CHAPTER 6

CONCLUSION & FUTURE WORK

6.1 Conclusion

This project successfully demonstrates that student Mathematics exam performance can be predicted with high accuracy ($R^2 = 0.869$) using a set of seven demographic and academic features. The Random Forest Regressor outperformed all other tested models and was deployed as a live web application on AWS Elastic Beanstalk.

The most significant finding is that reading and writing scores together account for over 84% of the model's predictive power, establishing cross-subject literacy as the strongest academic predictor. Structured test preparation course completion offers an actionable lever for institutions: students who completed the course scored meaningfully higher on average.

The deployed web interface makes the model's predictions accessible to educators without any programming knowledge, fulfilling the project's goal of building a practically usable tool. The system demonstrates how machine learning can be responsibly integrated into educational decision support, focusing on enabling timely intervention rather than high-stakes automated decisions.

6.2 Future Work

LMS Integration: Connect the prediction system to real-time Learning Management System data such as Moodle or Canvas, incorporating assignment submission rates, forum activity, and video engagement metrics as additional predictive features.

Multi-subject Prediction: Extend the model to simultaneously predict Reading and Writing scores in addition to Math, enabling a comprehensive academic profile prediction for each student.

SHAP-based Explanations: Integrate SHAP (SHapley Additive exPlanations) into the web application to provide per-student explanations of each prediction, increasing educator trust and actionability.

Student Dashboard: Build a secure student-facing portal where students can input their mid-semester data and receive feedback on their projected performance along with personalized study recommendations.

Automated Retraining Pipeline: Implement a CI/CD-style pipeline that automatically retrains the model each academic year using new student data, monitoring for model drift and automatically selecting the best new model.

Mobile Application: Develop an Android/iOS companion app for the web service, enabling educators to access predictions on the go and receive push notifications for at-risk student alerts.

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